

The background is a dark navy blue. On the left, there are two overlapping geometric shapes: a blue parallelogram and a light green parallelogram. Below these, there is a circular inset showing a close-up of a printed circuit board (PCB) with various electronic components. In the top right corner, there is a faint, stylized pattern of white lines resembling a circuit or a city map.

Interaction with **Team Flyby**

Speaker : Johaan

Meeting Agenda

Introductions

Technical Difficulties

Technical Questions

Competition Related Queries

Final thoughts and conclusions





Introductions

Team Name

SINGULARIT

Composition

30 Engineering undergraduates from Cochin University of Science and Technology, India.

Previous Experience

Placed 25th out of 49 teams in SUAS 2025.

Current Projects

Attempting the Mathworks Minidrone Competition and A2RL.



Technical Difficulties

We would love to hear about the technical difficulties
your team faced during development.

**What felt different about this competition compared
to others?**

Speaker : Safwaan



Technical Doubts and Questions

While preparing and designing the architecture for this project, we encountered some doubts and questions that we would like to ask your team.





Q. Were there any computation bottlenecks while using NVIDIA Jetson when the drone was at top speeds? What were the sensor modalities you are given to work with (anything other than CMOS camera & IMU)?





Q. Can you tell us more about your approach on state estimation and perception using available sensor modalities? How did you implement gate segmentation and localisation?



Speaker : Aaromal A



Q. How did you prepare your vision pipeline to survive severe indoor visual artifacts/noises such as similar gates, motion blur causing gate misclassifications?



Q. The standard Arducam IMX219 camera exhibits severe rolling shutter shearing during high-speed rotations. How was the team able to get around this?



Speaker : Adithya



Q. How did the team approach drone calibration between runs and after a crash? How did your EKF handle the transition period after a crash when calibration was incorrect?

how sensitive was your state estimator to small calibration residuals?

Q. How did the team handle 16g IMU saturation fallback logic during aggressive maneuvers like the “Split-S”?



Speaker : Ivana



Q. Tell us more about your planning & control architecture and latency issues you faced there?

Time-optimal geometric paths tracked by an MPC or did you use an RL policy to map your state estimates directly to CTBR commands?



Speaker : Sanjay



Q. How was opponent/collision detection implemented in the Silver Multi-Drone Race?



Speaker : Amaan



Q. How did you balance Jetson compute with low-level flight controller firmware (betaflight)? CTBR + Betaflight Mapping vs Direct RPM mapping

How do you address/design for the control latency at high speeds?

Speaker : Nandagopal



Q. How did your agent adapt its flight path to handle the drop in maximum motor RPM over time?

Speaker : Ivana



Q. Were there any software parameter adjustments to be made while shifting from simulation to real life testing? How you adapt RL policy to work in real environment?



Speaker : Amaan



Competition related Queries

We would love to hear about your
experience during the competition

Speaker : Gabriela



Q1. A summary of the full selection process — initial application, followed by simulation rounds or qualification stages, before the team was confirmed for the live event?

We primarily want to know about the competition timeline, qualification dates and deadlines to submit a registration.





Q2. What was your initial video submission based on — was it footage from a drone you had already built? What elements did it contain?

We are currently editing our own video, taken from our mission run at SUAS. Do you have any suggestions on how to make the video?





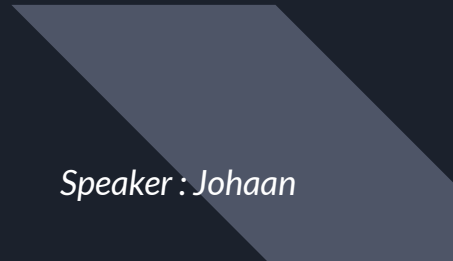
Q3. Was there any financial support provided by the organizers? In the form of hotel rooms or flights?



Speaker : Nandagopal



Q4. Was the whole experience enriching for the team? How was the reception at the venue and were there opportunities to network and collaborate?



Speaker : Johaan



Do you have any questions for us?



Thank you!

For taking time out of your busy
schedule for us, we are truly grateful
for Team Flyby!